12TH CHEMISTRY · CHAPTER 27 · FULL CHAPTER NOTES

Biochemistry

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] What is biochemistry? Name the essential biomolecules.

Biochemistry is the branch of science that deals with the study of chemical substances found in living organisms and the reactions they undergo. Biological compounds are complex molecules with the same types of functional groups as those found in simple organic compounds, such as alcohol, amine, aldehyde, ketone, carboxylic acid, ester and amide groups. The essential biomolecules are carbohydrates, proteins, lipids and nucleic acids; minerals and enzymes also take part in biochemical processes.

27.1 CARBOHYDRATES

Q.2 [Short] What are carbohydrates? How are they defined chemically?

Carbohydrates are naturally occurring organic compounds including sugars, starch, cellulose and substances found in bacterial cell walls and insect exoskeletons. They act as major sources of energy in our diet. All carbohydrates are hydrates of carbon. Chemically, they are polyhydroxy aldehydes or ketones or their derived compounds. They are classified as monosaccharides, disaccharides and polysaccharides.

Q.3 [Long] What are monosaccharides? How are they classified?

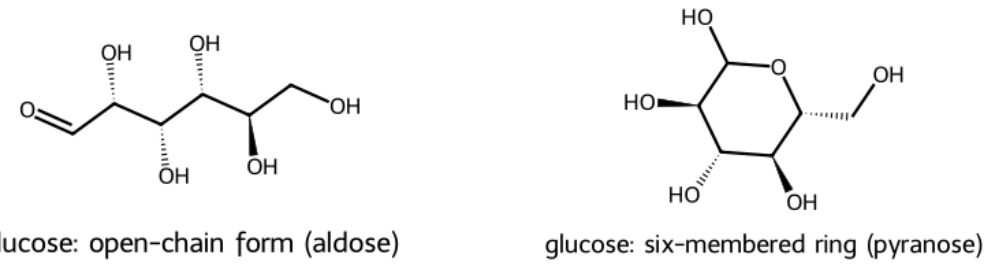
Monosaccharides are the simplest carbohydrates, commonly known as sugars, e.g. glucose and fructose. They cannot be hydrolysed further into simpler compounds. Their general formula is $C_n(H_2O)_n$, where n ranges from three to seven.

By number of carbon atoms: triose (3 C), tetrose (4 C), pentose (5 C), hexose (6 C), etc. **By functional group:** an aldose contains an aldehydic group and a ketose contains a ketonic group. Glucose is an aldose whereas fructose is a ketose.

The textbook sentence gives “glucose and sucrose” as examples of monosaccharides; sucrose is a disaccharide (the book says so itself under disaccharides), so fructose is written here.

Q.4 [Long] Describe the open-chain and ring structures of monosaccharides.

Monosaccharides exist either as open chains or as ring-shaped molecules. Pentoses and hexoses are more stable, and hence more abundant, as cyclic structures than as open-chain structures. According to the textbook, in the solid form they exist in the ring form and in solution their molecules exist as linear structures. Fructose forms a five-membered ring known as **furanose**, whereas glucose forms a six-membered ring known as **pyranose**. The linear and cyclic structures are interconvertible.



Linear and cyclic structures of glucose

Q.5 [Short] Why are monosaccharides solids and highly soluble in water?

The many hydroxyl groups (–OH) in sugar molecules give extensive hydrogen bonding between their molecules; for this reason monosaccharides are solids in their pure form. In solution they form extensive hydrogen bonds with water molecules, which makes them highly soluble in water. Monosaccharides are an excellent source of instant energy. Glucose is called blood sugar because it is present in human blood, while fructose is abundant in fruits and honey.

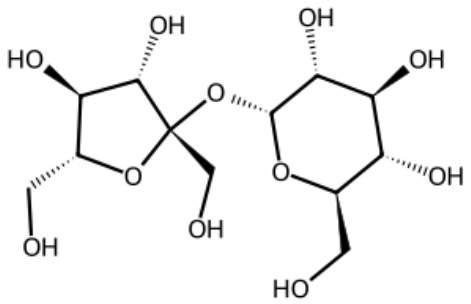
Q.6 [Short] Why is honey thick and viscous?

Fructose and glucose are present in honey in high concentrations. The thickness and viscosity of honey is due to their strong and intensive hydrogen bonding.

Q.7 [Long] What are disaccharides? How is the glycosidic bond formed?

Disaccharides consist of two monosaccharide units joined together by a glycosidic linkage; their general formula is $C_{12}H_{22}O_{11}$. The most common disaccharides are sucrose, lactose and maltose. They are formed by a condensation reaction: the hydroxyl group of one monosaccharide combines with the hydrogen of another, forming a covalent bond and releasing a molecule of water. The C–O–C bond is known as the **glycosidic bond**. For example, sucrose (common table sugar) is a disaccharide of glucose and fructose.

$$C_6H_{12}O_6 \text{ (glucose)} + C_6H_{12}O_6 \text{ (fructose)} \rightarrow C_{12}H_{22}O_{11} \text{ (sucrose)} + H_2O$$



Glycosidic linkage in sucrose

Q.8 [Short] What are polysaccharides?

Polysaccharides are large molecules consisting of a large number of monosaccharide units joined together. They have high molecular mass and the general formula $(C_6H_{10}O_5)_n$. On hydrolysis they give simple sugars (monosaccharides). Common examples are starch, glycogen, cellulose and pectin.

Q.9 [Long] Describe the structures of starch, cellulose and glycogen.

Starch is the main storage of energy in plants and consists of glucose units. It has two components. **Amylose** has long, unbranched chains with many (1,4) glycosidic linkages. **Amylopectin** is a compact, highly branched structure of many glucose units (1,4 links with 1,6 links at the branches). According to the textbook, amylopectin is soluble in water while amylose is insoluble. Potato, sugar beet, rice and grains contain large quantities of starch.

In general, amylose is the component that dissolves in hot water and amylopectin is the less soluble one; the statement above is written as the textbook gives it.

Cellulose is the major structural material of plant cell walls. It consists of glucose units (up to 2500) forming long unbranched chains linked by glycosidic bonds. These chains form strong hydrogen bonds with each other, making cellulose rigid and insoluble in water. It makes up most of the leaves, branches and trunk of plants. It is very important in human food in the form of fibre, which improves gut health.

Glycogen, or 'animal starch', is a polymer of glucose that is stored in the liver and muscles of animals. Its structure is very similar to that of amylopectin, but it is more highly branched, which makes it different from starch.

Polysaccharide	Structure	Function
Starch – amylose	Long, unbranched chains of glucose	Energy store of plants
Starch – amylopectin	Compact, highly branched chains of glucose	Energy store of plants
Cellulose	Long unbranched chains (up to 2500 glucose units) held together by strong hydrogen bonds	Structural material of plant cell walls; dietary fibre
Glycogen	Like amylopectin but more highly branched	Energy store of animals (liver and muscles)

Q.10 [Short] Why is cellulose rigid and insoluble in water?

Its long unbranched chains of glucose form strong hydrogen bonds with each other, which hold the chains firmly together; this makes cellulose rigid and insoluble in water.

Q.11 [Long] Describe the role of glucose in the body. What is the normal blood glucose level?

Simple sugars like glucose are the primary source of energy for cells. They are easily absorbed and metabolised to release energy; 1 g of glucose provides about 15.6 kJ. Excess glucose is stored as glycogen in the liver and muscles. Normally the blood glucose level ranges between 70–110 mg/dL and is controlled by the hormones insulin and glucagon. Insufficient carbohydrate intake can result in fatigue and reduced physical and mental performance.

Q.12 [Short] What is diabetes? What is hyperglycemia?

If the blood glucose level rises above the normal range, it can lead to diabetes, a chronic condition that occurs when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces. Uncontrolled diabetes can cause hyperglycemia (high blood sugar), which over time may damage body systems, particularly nerves and blood vessels.

Q.13 [Short] What are the effects of a high intake of fructose and refined sugars?

The main function of fructose is to provide energy during metabolism. Its high intake causes obesity, an excessive accumulation of fat in the body. Diets high in refined sugars are associated with a higher risk of chronic diseases such as diabetes, heart disease and certain cancers. Conversely, complex carbohydrates and fibres are protective, with multiple health benefits. When the body has sufficient carbohydrates for its energy needs, it uses these instead of breaking down proteins or fats for fuel.

Q.14 [Short] Why do animals and humans have large glycogen deposits?

Excess glucose is stored as glycogen in the liver and muscles. Glycogen is highly branched, so glucose units can be released quickly from its many ends whenever energy is needed. These large deposits allow animals and humans to carry on sustained muscular activity.

Q.15 [Short] Which complex carbohydrates lubricate the elbow and knee?

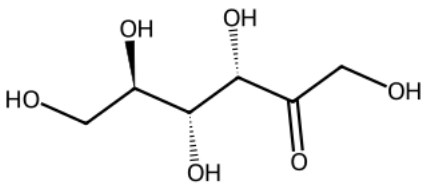
Complex carbohydrates (polysaccharides) such as **hyaluronic acid**, present in the synovial fluid of joints, provide lubrication to the elbow and knee.

This point is listed in the learning outcomes but is not explained in the chapter text; the name of the carbohydrate is given from general knowledge.

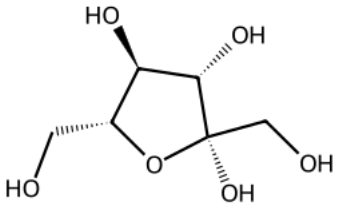
Quick Check 27.1

Q.16 [Short] a) The linear structure of fructose is given here. i. Show the conversion of this structure into the cyclic structure. ii. Predict whether fructose is soluble in water or not. Explain your reasoning. iii. Predict the physical state of fructose in its pure form by analyzing its structure.

i. The –OH group on carbon 5 adds to the ketonic C=O group on carbon 2, closing a five-membered ring (furanose).



fructose: open-chain form (ketose)



fructose: five-membered ring (furanose)

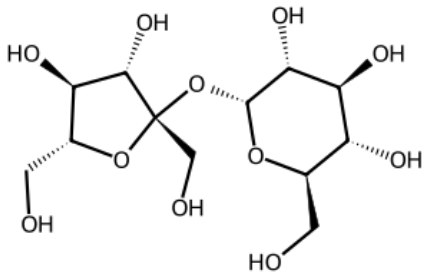
ii. Fructose is soluble in water. It has many hydroxyl (–OH) groups which form extensive hydrogen bonds with water molecules.

iii. It is a solid in its pure form, because the many –OH groups give extensive hydrogen bonding between its molecules.

Q.17 [Short] b) What is glycosidic linkage? Show how it is formed between fructose and glucose.

The C–O–C bond which joins two monosaccharide units is called the glycosidic linkage. It is formed by a condensation reaction: the hydroxyl group of glucose combines with the hydrogen of a hydroxyl group of fructose, a molecule of water is released and the two rings are joined through an oxygen atom to give sucrose.

$$C_6H_{12}O_6 \text{ (glucose)} + C_6H_{12}O_6 \text{ (fructose)} \rightarrow C_{12}H_{22}O_{11} \text{ (sucrose)} + H_2O$$



sucrose: glucose ring and fructose ring joined by a C–O–C glycosidic linkage

Q.18 [Short] c) Briefly describe any two functions of carbohydrates and how these functions are related to their molecular structures.

1. Energy storage: starch (plants) and glycogen (animals) store energy. They are polymers of glucose; their branched, compact structures pack many glucose units, which are released on hydrolysis.

2. Structural support: cellulose forms plant cell walls. Its long unbranched chains of glucose form strong hydrogen bonds with each other, making it rigid and insoluble in water.

27.2 PROTEINS

Q.19 [Short] What are proteins? What do they give on hydrolysis?

Proteins are large, complex macromolecules found abundantly in all living organisms. Our skin, hair, haemoglobin in blood (red pigment) and muscles are all proteins. They play key roles in the human body, from building tissue structure to various functions of the cells and organs. When proteins are completely hydrolysed they yield amino acids. About 20 different amino acids link together in various sequences to create a wide variety of proteins.

Q.20 [Short] What is a peptide bond? What are polypeptide chains?

Proteins are linear chains of amino acids linked together by peptide bonds. A peptide bond is an amide bond that forms when the –COOH group of one amino acid reacts with the $–NH_2$ group of the adjacent amino acid, with loss of water. The long chains of amino acids formed by many peptide linkages are called polypeptide chains.

$$H_2N–CHR–COOH + H_2N–CHR'–COOH \rightarrow H_2N–CHR–CO–NH–CHR'–COOH + H_2O$$

Q.21 [Long] Describe the four structural levels of proteins.

Primary structure: the specific linear sequence of amino acids linked by peptide bonds in a polypeptide chain. This sequence determines how the protein folds into its three-dimensional shape, which in turn defines its biological function. Even a small change in a single amino acid can alter the structure and impair the function. Proteins like albumin (about 400 amino acids) and insulin rely on their precise amino acid sequences to function properly.

Secondary structure: it arises when the backbone atoms form hydrogen bonds within a polypeptide chain, producing regular patterns such as α-helices and β-pleated sheets. In an **α-helix**, hydrogen bonds form along a single chain, creating a coiled, spiral structure (e.g. keratin). **β-pleated sheets** form when hydrogen bonds occur between adjacent polypeptide strands, giving a folded, sheet-like structure (e.g. silk fibroin). These arrangements provide stability and flexibility.

Tertiary structure: the overall three-dimensional shape formed when a polypeptide chain folds due to interactions between the side chains (R-groups): hydrogen bonds, ionic bonds, hydrophobic interactions and disulphide bridges. These interactions make the chain twist and bend into a specific shape, often forming globular proteins. This three-dimensional structure determines the function of the protein. For example, in enzymes the precise shape of the active site allows specific binding of the substrate; any change in this structure can affect the enzyme activity.

Quaternary structure: some proteins have it too. It arises when two or more folded polypeptide chains (subunits) associate by interactions such as hydrogen bonds, ionic (salt) bridges, hydrophobic interactions and sometimes disulphide bonds.

Level	What it is	Bonds involved	Example / function
Primary	Linear sequence of amino acids	Peptide bonds	Decides folding and function (albumin, insulin)
Secondary	α-helix or β-pleated sheet	Hydrogen bonds of the backbone	Keratin (helix), silk fibroin (sheet)
Tertiary	Overall 3-D folding of one chain	H-bonds, ionic bonds, hydrophobic interactions, disulphide bridges	Active site of enzymes
Quaternary	Two or more chains (subunits) together	Same interactions between subunits	Proteins with several subunits

Q.22 [Short] What is sickle cell anaemia? What does it show about primary structure?

Sickle cell anaemia is a disease caused by the deformation of normal red blood cells into cells having a sickle-like shape. It is caused by the abnormal formation of haemoglobin (a change in the amino acid sequence). Sickle cells clog the blood veins and disrupt the normal blood circulation.

Q.23 [Long] Differentiate between fibrous and globular proteins.

Fibrous proteins	Globular proteins
Long, rod-like or thread-like molecules; polypeptide chains run parallel.	The whole molecule is spherical or ovoid.
Chains are linked by strong intermolecular hydrogen bonds or disulphide bonds to form fibres.	Chains are folded into a compact shape.
Generally insoluble in water, mechanically strong and stable.	More sensitive to heat and pH (easily denatured).
They serve as structural polymers.	Most worker (functional) proteins: enzymes, hormones and nutrients.
Collagen (connective tissue), keratin (hair, nails, skin), myosin (muscle); silk fibroin.	Egg white (albumin), casein of milk, insulin.

Q.24 [Short] What is denaturing of proteins? Why does lemon juice precipitate milk protein?

Globular proteins are sensitive to heat and pH. The white of an egg hardens on heating because the hydrogen bonding between the chains is disrupted and the chains open up. For the same reason the milk protein **casein** is coagulated (precipitated) when the pH of the solution is lowered by adding some lemon juice. This is called **denaturing** of proteins.

Q.25 [Short] Describe the fibrous proteins of hair and silk.

Hair contains **keratin**, a fibrous protein whose chains are coiled as α-helices and linked by hydrogen bonds and disulphide bonds; it gives strength and flexibility. Silk contains **fibroin**, a fibrous protein made of β-pleated sheets in which adjacent strands are held by hydrogen bonds. Both are insoluble in water, mechanically strong and stable.

Q.26 [Long] Classify proteins according to their biological functions.

Type of protein	Function	Example
Hormonal	Control or regulate growth, development, metabolism and reproduction	Insulin (from the pancreas) regulates movement of glucose from blood into cells
Storage	Provide nutrients for germinating seedlings or early development of animals	Ferritin stores iron
Structural	Fibrous; maintain shape and give support; form the matrix of skin, ligaments, tendons and bones	Keratin in hair, fibrin in blood clots, collagen
Contractile	Thin and thick filaments of muscle; contraction and relaxation	Actin and myosin
Defensive	Fight infection and prevent injury	Antibodies bind to specific antigens
Transport (carrier)	Move molecules and ions across membranes and around the body	Haemoglobin carries oxygen on its iron atom
Receptor	Located in the cell membrane; receive chemical signals	—
Regulatory	Control cellular and physiological activities	Insulin, growth hormone

Q.27 [Short] What is insulin? How is it related to diabetes mellitus?

Insulin is a protein hormone produced by the pancreas. It regulates the movement of glucose from the blood into the cells (sugar metabolism). When the pancreas does not produce enough insulin, or the body cannot use it effectively, the blood glucose level rises above the normal range (70–110 mg/dL); this condition is diabetes mellitus.

Q.28 [Short] How do cholesterol and amino acids serve as hormones?

Cholesterol is a steroid; steroid hormones are among the steroids of biological systems and are made from it. **Amino acids** are the units of protein hormones such as insulin and growth hormone, which regulate sugar metabolism and growth.

The learning outcome asks how cholesterol and amino acids serve as hormones; the chapter text gives only these points.

Quick Check 27.2

Q.29 [Short] **a) Keratin is a structural protein found in hair, nails, horn, hoofs, wool, feathers, enamel of teeth and outer layer of skin, where it gives strength and flexibility. It is not easily denatured. Which bond do you expect is responsible for this; hydrogen bond or disulphide bond? Explain.**

The **disulphide bond**. It is a strong covalent bond (–S–S–) between the chains and is not easily broken by heat or a change of pH, whereas hydrogen bonds are weak and are easily disrupted. So the disulphide bridges keep the keratin chains firmly linked.

Q.30 [Short] **b) How can two peptides with identical type and number of amino acids differ in their primary structure?**

The primary structure is the specific linear **sequence** of amino acids. The same amino acids can be joined in a different order (e.g. Gly–Ala–Ser and Ala–Gly–Ser), which gives different primary structures and therefore different peptides.

Q.31 [Short] **c) What are the two most common types of secondary protein structures in our body?**

The α-helix and the β-pleated sheet.

Q.32 [Short] **d) Why does a denatured protein no longer function properly?**

The function of a protein depends on its specific three-dimensional shape. On denaturation the hydrogen bonds (and other interactions) that hold this shape are disrupted and the chains open up, so the shape (e.g. the active site of an enzyme) is lost and the protein cannot work.

Q.33 [Short] **e) The white of egg solidifies on heating. Explain why it happens.**

Egg white is a globular protein. On heating, the hydrogen bonding between the chains is disrupted and the chains open up, so the protein is denatured and hardens (coagulates).

27.3 ENZYMES AS BIOCATALYSTS

Q.34 [Short] **What are enzymes? Give an example showing their efficiency.**

Enzymes are proteins that act as biological catalysts for the chemical reactions that take place in living organisms. An enzyme speeds up a biochemical reaction to an extent that would be impossible without it. For example, one molecule of carbonic anhydrase in the blood can convert about one million carbon dioxide and water molecules into carbonic acid in one minute.

Q.35 [Long] **Describe the mechanism of enzyme action and the lock and key model.**

An enzyme has a special site on its surface, called the **active site**, which binds the substrate molecule and converts it into the products. The random movement of enzyme and substrate molecules allows the substrate to enter the active site. A temporary **enzyme–substrate complex** is formed. After this, the products are released and the enzyme remains unchanged, free to bind another substrate molecule.

$$\text{Enzyme} + \text{Substrate} \rightleftharpoons \text{Enzyme–substrate complex} \rightarrow \text{Enzyme} + \text{Products}$$

Lock and key model (Emil Fischer, 1894): the lock is the enzyme and the key is the substrate. Only the correctly shaped key (substrate) fits into the keyhole (active site) of the lock (enzyme).

Q.36 [Short] Why are enzymes specific in their action?

Enzymes are highly specific: each enzyme acts on only one particular substrate. This specificity is due to the active site of the enzyme, which has a unique shape that allows only a specific substrate to fit into it, like a key in a lock.

Q.37 [Long] Describe the factors affecting enzyme activity.

The activity of an enzyme is influenced by temperature, pH and the concentrations of substrate and enzyme. It may also be affected by other substances such as co-enzymes, activators and inhibitors.

Temperature and pH: the temperature and pH that give maximum enzyme activity are called the optimum temperature and optimum pH. If they are disturbed from the optimum values, the enzyme can be deactivated. Enzymatic reactions in the human body occur best at or around 37 °C and in a pH range of 6.4–6.9.

Concentration of enzyme and substrate: the rate increases as the enzyme or substrate concentration is increased, because more enzyme–substrate complex is formed. However, increasing the substrate concentration beyond a limit can slow down the reaction.

Q.38 [Short] What are enzyme inhibitors?

A variety of small molecules or ions that cause enzymes to lose their catalytic activity are called inhibitors. Different inhibitors behave differently, but they all prevent the active site from binding with a substrate.

Q.39 [Long] Explain competitive and non-competitive inhibition with examples.

Reversible (competitive) inhibitors are molecules having the same shape as the substrate. The substrate and the inhibitor compete for the active site of the enzyme; the more the inhibitor, the more the enzyme is deactivated for the original substrate. Example: butanedioate (succinate ion) is oxidised by the enzyme succinate dehydrogenase; the propanedioate ion (malonate) inhibits this enzyme because it resembles the substrate.

$$^{-}OOC–CH_2–CH_2–COO^{-}\ \text{(succinate ion)}\quad ^{-}OOC–CH_2–COO^{-}\ \text{(malonate ion)}$$

Non-competitive (irreversible) inhibitors are small ions or molecules that change the shape of the active site so that it is no longer available for the substrate. Such inhibition is irreversible and sometimes causes severe health issues or even death. An enzyme to which an irreversible inhibitor (poisons, antibiotics, anti-metabolites and some drugs) is attached loses its activity permanently. Example: penicillin is an irreversible inhibitor that blocks the enzyme trans-peptidase, which is needed for the synthesis of the bacterial cell wall; thus it kills bacteria by disrupting the cell wall.

The textbook sentence reads "Thus insulin kills bacteria..."; the antibiotic being described is penicillin.

Competitive inhibition	Non-competitive inhibition
The inhibitor has the same shape as the substrate.	The inhibitor is a small ion or molecule that need not resemble the substrate.
It occupies the active site and competes with the substrate.	It changes the shape of the active site so that the substrate cannot bind.
Reversible.	Irreversible; activity is lost permanently.
Example: malonate inhibits succinate dehydrogenase.	Example: penicillin blocks trans-peptidase; poisons.

Q.40 [Short] How does cobra venom cause death?

Cobra venom has an enzyme, PLA2, which can disrupt the cell wall of the red blood cell. This results in hemolysis, which causes swelling of the affected organ and ultimately death.

Quick Check 27.3

Q.41 [Short] a) How do enzymes differ from inorganic catalysts?

Enzymes are proteins (biological catalysts) and are highly specific: each acts on only one particular substrate, which fits its active site. They work best only at an optimum temperature (about 37 °C) and pH, and are deactivated outside these. Inorganic catalysts are not proteins, are less specific and can work over a wide range of temperature and pH.

Q.42 [Short] b) What is the effect of decreasing and increasing temperature affect an enzyme activity?

Enzyme activity is maximum at the optimum temperature (about 37 °C in the human body). On decreasing the temperature the molecules move more slowly, fewer enzyme–substrate complexes form and the activity falls. On increasing the temperature above the optimum the enzyme (a protein) is denatured and its action is deactivated.

Q.43 [Short] c) Differentiate between competitive and non-competitive inhibitors.

Competitive inhibition	Non-competitive inhibition
The inhibitor has the same shape as the substrate.	The inhibitor is a small ion or molecule that need not resemble the substrate.
It occupies the active site and competes with the substrate.	It changes the shape of the active site so that the substrate cannot bind.
Reversible.	Irreversible; activity is lost permanently.
Example: malonate inhibits succinate dehydrogenase.	Example: penicillin blocks trans-peptidase; poisons.

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Q.44 [Short] **d) An enzyme is a protein. When the pH is changed, the function of the enzyme is disturbed. In terms of the protein structure, explain how and why it happens so?**

The active site has a precise shape which is part of the tertiary structure, held by hydrogen bonds and ionic bonds between the R-groups. A change in pH changes the charges on these groups and disrupts these bonds. The chain unfolds (denatures), the shape of the active site changes and the substrate can no longer bind.

27.4 LIPIDS

Q.45 [Short] **What are lipids? How are they classified?**

Lipids are a class of organic compounds found in animals and plants that include fats and phospholipids (such as those in the cell membrane), steroids and terpenes (essential oils). Unlike polysaccharides and proteins, lipids are not polymers: they lack a repeating monomeric unit. They are hydrophobic but soluble in organic solvents. They serve as fuel molecules, signal molecules, components of membranes, hormones and intracellular messengers. Based on chemical composition, lipids are classified into simple lipids, compound (complex) lipids and derived lipids.

Q.46 [Long] **What are simple lipids? Describe triglycerides.**

Simple lipids are esters of fatty acids with different types of alcohols; they are sub-classified by the type of alcohol into fats, oils and waxes.

Triglycerides: animal fats and vegetable oils are esters formed from one glycerol molecule and three fatty acids, which may be similar or different. This structure makes them efficient energy storage molecules; they also provide insulation, protection of organs and a source of fat-soluble vitamins.

$$CH_2OH–CHOH–CH_2OH + 3\ R–COOH \rightarrow CH_2(OOCR)–CH(OOCR)–CH_2(OOCR) + 3\ H_2O$$

glycerol + three fatty acids → triglyceride + water

Q.47 [Short] **Differentiate between fats and oils.**

Fats	Oils
Contain saturated fatty acids (no C=C double bonds).	Contain unsaturated fatty acids with one or more C=C double bonds.
Solid at room temperature.	Liquid at room temperature.
Mostly of animal origin (animal fats).	Mostly of plant origin (vegetable oils).

Q.48 [Short] Why are fats insoluble in water? How is this useful to the body?

The hydrophobic nature of the fatty acid chains makes fats insoluble in water but soluble in organic solvents. In the body this property allows them to form insulating layers under the skin and cushioning around organs such as the kidneys, helping in temperature regulation and protection.

Q.49 [Short] What are waxes? Give their functions and uses.

Most waxes are made up of long fatty acid chains esterified to long-chain alcohols. Both chains are non-polar and hydrophobic. Because of this, waxes serve as barriers by forming protective coatings which prevent water from sticking to the surface. Natural examples include beeswax, the oil in a whale's head, the feathers of some aquatic birds and plant waxes. They are widely used in the pharmaceutical, cosmetic and other industries to make lotions, ointments and polishes.

Q.50 [Long] What are compound lipids? Describe phospholipids, lipoproteins and glycolipids.

Compound lipids are esters of fatty acids with alcohols containing an additional group, such as phosphate, nitrogenous base, carbohydrate or protein.

Phospholipids: similar to a fat molecule but with only two fatty acids attached to glycerol; the third hydroxyl group is linked to a phosphate group, which makes the molecule polar. The hydrocarbon tails are hydrophobic, while the phosphate group and its attached components form a hydrophilic head. In water, phospholipids form a double-layered sheet called a **bilayer**: the polar ends face outward and the non-polar ends cluster together inside. It forms the structure of all cell membranes.

Lipoproteins: complexes of lipids and proteins that transport hydrophobic lipids (like cholesterol and triglycerides) in the bloodstream.

Glycolipids: lipids with carbohydrate (sugar) groups attached, mostly found in nervous tissue.

Q.51 [Short] What are liposomes? Why are they used for drug delivery?

Liposomes are artificial phospholipid structures used for the delivery of hydrophilic drugs. They are ideal for delivering water-loving cancer drugs because their structure, a lipid bilayer surrounding a watery core, allows them to encapsulate water-soluble agents, shielding them from degradation and enabling cellular uptake. They improve solubility and provide targeted delivery to the tumour.

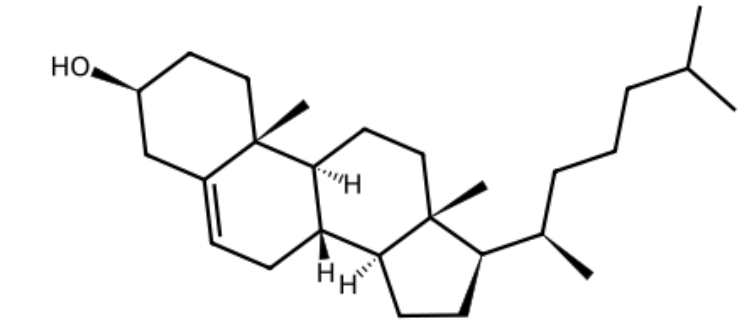
Q.52 [Short] What are derived lipids?

Derived (associated) lipids are the hydrolytic products of simple and compound lipids which have the same characteristics as lipids, e.g. fatty acids, glycerol, steroids and lipid-soluble vitamins. They are not fats themselves, but come from fats. They have important roles in energy and hormone production, cell structure, nerve health and vitamin function.

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Q.53 [Long] Describe the structure of steroids.

Steroids are not formed from fatty acids but have some lipid-like properties. They consist of three cyclohexane rings and one cyclopentane ring fused together, called the **steroid nucleus**; the four rings are named A, B, C and D. Many steroids also have the –OH functional group, which puts them in the alcohol class (sterols). The steroids of biological systems include cholesterol, bile salts, vitamin D and steroid hormones.



cholesterol: steroid nucleus (three six-membered rings + one five-membered ring) with an -OH group

Cholesterol, a steroid with the four fused rings A–D

Q.54 [Short] What are terpenes and carotenoids?

Lipids present in plants include a wide variety of hydrocarbons called terpenes, which are built from isoprene units. These hydrocarbons and their oxygenated derivatives have fewer than 40 carbon atoms. Terpenes that contain 40 carbon atoms are known as **carotenoids**. Example: limonene is a terpene in lemon.

$CH_2=C(CH_3)–CH=CH_2$ (isoprene)

Q.55 [Long] Describe the nutritional and biological importance of lipids.

- **Energy storage:** lipids are a primary store of energy. They provide more energy per gram than carbohydrates because they have a higher caloric value when oxidised to carbon dioxide and water.
- **Transport of vitamins:** they act as solvents for the fat-soluble vitamins (A, D, E and K) and help in their absorption in the intestinal cells and later into the bloodstream; these vitamins should be eaten with a bit of healthy fat.
- **Insulation:** fats insulate the body from water, heat and electricity due to their non-polar nature; they also cushion organs such as the kidneys.
- **Structure:** phospholipids form the bilayer of all cell membranes; steroids give hormones, bile salts and vitamin D.

Q.56 [Short] Why do hibernating animals accumulate fat?

Animals such as polar bears, reptiles and amphibians store extra fat when hibernating. Hibernation is a condition in which animals significantly slow down their metabolic rate, body temperature, heart rate and breathing to save energy for periods when food is scarce. Fat is a more concentrated source of energy than glycogen, which allows the animals to use stored fat effectively while minimising energy consumption.

Quick Check 27.3 (second box, lipids)

Q.57 [Short] a) Why lipids are not soluble in water?

Because of the long, non-polar (hydrophobic) hydrocarbon chains of their fatty acids, which cannot form hydrogen bonds with water. They dissolve in organic solvents instead.

Q.58 [Short] b) What are three major functions of lipids in the human body?

(1) They are a primary store of energy. (2) They act as solvents for the fat-soluble vitamins (A, D, E and K) and help in their absorption. (3) They insulate the body and cushion organs such as the kidneys (phospholipids also form cell membranes).

Q.59 [Short] c) What type/s of lipid do/does not contain fatty acids? Give examples.

Steroids and terpenes are not formed from fatty acids. Examples: cholesterol, bile salts, vitamin D and steroid hormones (steroids); limonene and carotenoids (terpenes).

Q.60 [Short] d) Vegetable oil and vegetable ghee both are triglycerides. What is the major structural difference between the two?

Vegetable oil contains unsaturated fatty acids with one or more C=C double bonds and is a liquid. Vegetable ghee contains saturated fatty acids (no C=C double bonds) and is a solid at room temperature.

27.5 NUCLEIC ACIDS

Q.61 [Long] What are nucleic acids? What is a phosphodiester linkage?

Nucleic acids are large biomolecules that store and transmit genetic information. They are polymers made up of smaller units called **nucleotides**. They are formed from nucleotide chains containing alternating sugar and phosphate groups, with organic bases attached to the sugar molecules. The sugars are ribose in RNA and deoxyribose in DNA; both are five-membered sugars.

Nucleotides are joined together by a **phosphodiester linkage** between the 5′ and 3′ carbon atoms of the sugars. The process is repeated up to several million times to make a polynucleotide. Phosphodiester linkages are strong covalent bonds, which give strength and stability to the polynucleotide chain.

Q.62 [Long] How does DNA store genetic information?

DNA is present in the nucleus of every living cell. It is a coded plan (‘blueprint’) for making proteins. Each specific protein is made by a specific gene, a section of DNA. The genetic information of an organism is stored in DNA molecules, which carry the genetic instructions for the development, functioning, growth and reproduction of all living organisms. The information is stored in the sequence of its four nitrogenous bases; a group of three bases (a codon) codes for a specific amino acid. Genes are organised into larger units called chromosomes; humans receive 23 pairs from their parents.

Q.63 [Short] Why do different cells make different proteins although they have the same chromosomes?

Nearly every cell in the body carries the full set of 23 pairs of chromosomes, but each cell uses only the genes it needs: a red blood cell activates the gene for haemoglobin while keeping the gene for keratin switched off.

Q.64 [Long] Describe the Watson and Crick double helix structure of DNA.

In 1953, James Watson and Francis Crick showed that DNA consists of two polynucleotide chains. Each chain forms a right-handed helical spiral and the two chains coil around each other to form a **double helix**. The two chains are not identical but run opposite to each other: the 3′ end of one faces the 5′ end of the other. Each chain has a sugar–phosphate backbone with bases projecting at right angles, held by weak hydrogen bonds to the bases of the opposite chain.

Base pairing: adenine (A) is always opposite thymine (T), and guanine (G) is opposite cytosine (C). There are two hydrogen bonds in the A–T pair and three hydrogen bonds in the G–C pair.

Q.65 [Short] What are genes? What is replication?

Genes are the specific regions of DNA that carry genetic information; they provide instructions for the cell to make a particular protein. Replication is the process during the cell cycle in which DNA is copied, so that when the cell divides each new cell receives a complete set of genetic material. In this way the genetic information is transmitted from one cell to the daughter cells and to the next generation.

Q.66 [Long] Describe the structure and types of RNA.

RNA is usually single stranded, unlike DNA, but the chain can fold back on itself to form a helical loop. A ribonucleotide chain contains ribose, four nitrogenous bases (A, U, G and C) and the phosphate group. There are three major types of RNA:

Type of RNA	Function
Messenger RNA (mRNA)	Carries the genetic blueprint copied from the DNA in the nucleus.
Ribosomal RNA (rRNA)	Helps to make up the ribosome.
Transfer RNA (tRNA)	Translates the nucleotide sequence in mRNA into the corresponding sequence of amino acids in a protein.

Q.67 [Short] Differentiate between DNA and RNA.

DNA	RNA
Sugar is deoxyribose.	Sugar is ribose.
Bases: A, T, G and C.	Bases: A, U, G and C (uracil in place of thymine).
Double stranded (double helix).	Usually single stranded; can fold back into a helical loop.
Stores the genetic information.	Transfers the information for protein synthesis (mRNA, rRNA, tRNA).

Q.68 [Long] Describe protein synthesis (transcription and translation).

Protein synthesis transfers the genetic information encoded in DNA. Two processes, transcription and translation, are required.

Transcription: the genetic instructions for making a protein are copied from a DNA gene into mRNA in the nucleus. When the mRNA strand is complete, it detaches from DNA and enters the cytoplasm, where it attaches to ribosomes.

Translation: at the ribosomes, the code carried by mRNA is read and converted into a specific sequence of amino acids. Amino acids are arranged according to the sequence of bases in the RNA. The tRNA molecules release their amino acids, which join together in a chain; the polypeptide chain grows as bonds form between the amino acids, and finally the complete protein molecule is formed.

DNA → (transcription) → mRNA → (translation) → protein

Quick Check 27.4

Q.69 [Short] a) Compare the stability provided by the sugar-phosphate backbone versus the inter-chain base pairing. In your answer, refer to the specific types of bonds and the number of bonds formed between: Adenine (A) and Thymine (T); Guanine (G) and Cytosine (C).

The sugar–phosphate backbone is held by **phosphodiester linkages**, which are strong covalent bonds; they give strength and stability to each polynucleotide chain. The two chains are held to each other by **hydrogen bonds** between the bases, which are weak: **two** hydrogen bonds between A and T and **three** between G and C. So the backbone gives the greater stability, while the weak base pairing allows the strands to separate for replication and transcription.

Q.70 [Short] b) How the DNA and RNA are chemically different? What are the similarities between these two?

DNA	RNA
Sugar is deoxyribose.	Sugar is ribose.
Bases: A, T, G and C.	Bases: A, U, G and C (uracil in place of thymine).
Double stranded (double helix).	Usually single stranded; can fold back into a helical loop.
Stores the genetic information.	Transfers the information for protein synthesis (mRNA, rRNA, tRNA).

Similarities: both are polymers of nucleotides; both have a sugar–phosphate backbone joined by phosphodiester linkages; both contain the bases A, G and C; both carry genetic information.

Q.71 [Short] c) Nearly every cell in the human body contains the exact same 23 pairs of chromosomes, yet a red blood cell produces haemoglobin while a skin cell produces keratin. Explain why.

Each cell activates only the genes it needs and keeps the others switched off. A red blood cell activates the gene for haemoglobin while keeping the gene for keratin switched off; in a skin cell the keratin gene is active.

27.5 MINERALS

The textbook numbers both "Nucleic Acids" and "Minerals" as section 27.5.

Q.72 [Short] What are minerals? How are they classified?

Minerals are inorganic substances necessary for the maintenance of certain physicochemical life processes. Although they produce no energy, they have important functions in various activities. They are broadly classified as macro (major) or micro (trace) elements. According to the textbook, the macro-minerals include calcium, phosphorus, sodium and chloride, and are required in amounts greater than 100 mg/dL; the micro-elements include iron, copper, cobalt, potassium, magnesium, iodine, zinc, manganese, molybdenum, fluoride, chromium, selenium and sulfur, and are required in amounts less than 100 mg/dL.

Q.73 [Long] Give the sources and the role in nutrition of iron, calcium, phosphorus and zinc.

Mineral	Sources	Role in nutrition
Iron	Liver, meat, egg yolks, nuts, enriched or whole grains, beans, peas and lentils	Synthesis of haemoglobin (red blood cells) and myoglobin (muscle cells); needed to carry oxygen

Mineral	Sources	Role in nutrition
Calcium	Milk and other dairy products, greens, broccoli, salmon, sardines, beans, peas and lentils	Formation of bones and teeth, normal blood clotting, muscle contraction and relaxation, heart and nerve function
Phosphorus	Meat, fish, poultry, eggs, milk, cereal products	Formation of bones and teeth; regulates release and use of body energy (ATP); carries fat as part of phospholipids; maintains acid/base balance
Zinc	Meat, seafood, whole grains	Component of many enzyme systems (growth, immune function); found in the hormone insulin
Sodium, potassium and chloride	Salt, meat, seafood, fruit juices, vegetables, lentils	Fluid balance, nerve transmission and muscle function

Q.74 [Short] Which mineral is required for blood coagulation?

Calcium. It is needed for normal blood clotting, as well as for bones and teeth, muscle contraction and relaxation, and heart and nerve function.

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